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Comparison of Full Frequency Inversion, Constrained Sparse Spike Inversion and Geostatistical Inversion in Predicting Complex Carbonate Progradation Thin Reservoirs

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Abstract

Accurate characterization of thin, heterogeneous carbonate reservoirs remains a significant challenge due to complex sedimentary processes, limited vertical resolution, and the limitations of traditional seismic inversion techniques. This study illustrates the advantages of Full-Frequency Inversion (FFI) by comparing it with Constrained Sparse Spike Inversion (CSSI) and Geostatistical Seismic Inversion (GSI) in the prediction of progradational carbonate tidal channel reservoirs.

FFI integrates waveform-driven seismic analysis, facies-controlled modeling, and shaped-spectrum synthesis across four frequency bands to produce high-resolution impedance volumes. It significantly improves both vertical and lateral resolution, overcoming the stochastic nature of GSI and the limited frequency bandwidth of CSSI. The method enables reliable prediction of thin reservoirs down to 2–3 meters in thickness and supports high-fidelity simulation of non-impedance attributes such as resistivity, gamma ray, and porosity.

Applied to a Cretaceous carbonate formation in northern UAE, FFI successfully identified three main progradational bodies with distinct stacking and migration patterns. The results show good agreement with well data, accurately delineating high-permeability zones and improving the prediction of reservoir architecture. Compared to GSI, FFI achieved similar resolution with significantly shorter computation time and greater determinism.

This case study demonstrates that FFI is a robust and efficient tool for predicting thin, laterally variable carbonate reservoirs. Its integration of geological constraints and seismic waveform characteristics makes it especially valuable for development planning in complex carbonate settings.

The objective is to illustrate the advantages of full-frequency inversion (FFI) technology by comparing the application examples of full-frequency inversion, Constrained sparse spike inversion (CSSI) and geostatistical seismic inversion (GSI) in complex carbonate reservoir prediction. Full-frequency inversion technology can achieve accurate prediction of complex carbonate thin reservoirs and significantly improve the vertical resolution, lateral resolution and reservoir prediction accuracy. First, the principle of full-

frequency inversion is introduced, then FFI is used to predict carbonate progradation reservoirs. Secondly, CSSI and GSI are used to predict carbonate progradation reservoirs. Finally, the three inversion technologies are compared and analyzed through examples to illustrate the application advantages and effect comparison of full-frequency inversion technology in the prediction of complex carbonate thin layers.

Geological Setting and Technical Challenges

The study area is located in the northern UAE and features a typical carbonate platform setting formed during the Cretaceous. The targeted formation is approximately 100 meters thick and exhibits complex progradation sedimentation, characterized by lateral migration and vertical stacking due to multiple sedimentary cycles.

The study area is located in a carbonate platform margin setting (gentle to moderate slope) with complex sedimentary microfacies. Carbonate sedimentation is dominated by aggradation and progradation processes, resulting in facies of shoal stacking with complex spatial superposition relationships. The distribution pattern of the reservoirs is weakly defined, characterized by a series of progradational features and significant heterogeneity. Sweet spots are mainly distributed in high-energy depositional environments such as backshoals and grain shoals, as well as in areas with well-developed dissolution features.

Due to the complex pore networks and weak attribute sensitivity, quantitative relationships between porosity and permeability are weak, and seismic attributes often fail to effectively indicate sweet spots. Additionally, the development of thin high-permeability "thief zones" exacerbates water breakthrough and reduces sweep efficiency. Characterized by rapid vertical and lateral facies variation, the stacking and migration of progradational units produce rapid facies transitions over short distances. This undermines the reliability of conventional deterministic seismic inversion methods, which struggle to resolve sub-seismic-scale layering.

Existing inversion techniques have certain limitations. For instance, Sparse-spike inversion suffers from limited vertical resolution, while geostatistical inversion is heavily dependent on dense well control and suffers from strong model dependency and lateral randomness. These limitations motivate the adoption of a more robust and data-integrated approach such as full-frequency inversion (FFI).

The key advantage of Waveform-driven full-frequency inversion (FFI) is the full utilization of 3D seismic waveform features, aligning better with geological trends. Carbonate reservoirs, particularly those in the Middle East, are characterized by extreme heterogeneity, complex pore systems, and highly variable sedimentary environments. These factors often hinder accurate reservoir characterization and effective development planning. Traditional seismic inversion techniques, including sparse-spike and model-based inversion, are constrained by limited seismic bandwidth and oversimplified assumptions about reservoir continuity. Moreover, geostatistical inversion methods, though capable of simulating high-frequency components, frequently rely heavily on prior models and dense well coverage—conditions rarely met in carbonate settings.

In this paper, we integrate the geological concept of "facies control" into inversion, shifting the process from random simulation toward deterministic prediction. Reduced subjectivity and randomness is achieved, resulting in higher accuracy and reliability. FFI is applied to predict progradation layers as thin as 2–3 meters, significantly improving vertical and lateral resolution of reservoir prediction.

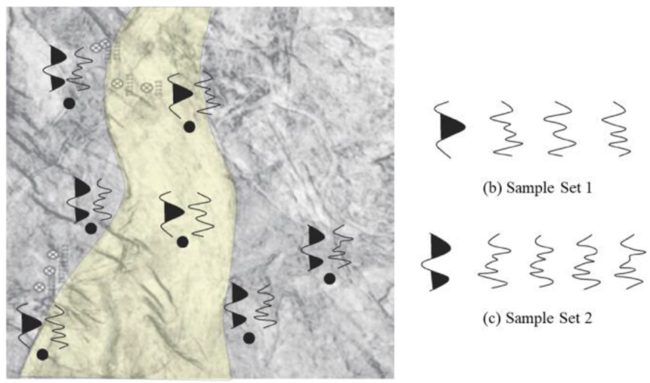
Three main technical challenges impact reservoir characterization in this area:

- Classical inversion theory cannot breakthrough the seismic resolution limit, leading to a high degree of ambiguity in thin reservoir prediction;
- Existing conventional inversion techniques rely heavily on well interpolation for high-frequency simulation, resulting in a strong stochastic nature in thin reservoir prediction;
- Traditional well-seismic combined inversion lacks constraints from geological context, leading to poor reliability in thin reservoir prediction.

Methodology: Full-Frequency Seismic Inversion (FFI)

Overview

Full-frequency inversion (FFI) is a seismic waveform-driven inversion method that aims to reconstruct a broadband impedance model by integrating multiple frequency bands—low, mid, high, and ultra-high frequencies—using a structured spectrum synthesis approach. The method combines seismic waveform analysis, statistical clustering, and Bayesian inversion frameworks to provide high-resolution predictions of subsurface properties. As figure 1 shows the seismic waveform is compared and analyzed with the well log bypass seismic track. Wells with similar waveform characteristics are selected as effective samples to construct a likelihood function that conforms to the seismic medium-frequency impedance and well curve structure characteristics.



(a) Seismic waveform and well logging curve characteristics

Figure 1—Basic schematic diagram of seismic waveform driven inversion

Characteristics of the Full-Frequency Inversion Method:

1. High Vertical and Lateral Resolution
   The full-frequency inversion method leverages lateral variations in seismic waveforms to drive high-frequency log features, achieving high-resolution inversion results. Vertically, the inversion aligns with high-frequency components from well logs, resulting in fine vertical resolution (when the common high-frequency structure in logs is around 200–300 Hz, the vertical resolution for thin-layer prediction can reach 2–3 meters). Laterally, the results conform to the lateral variation of seismic waveforms, offering equally high horizontal resolution. Thus, the method enhances both vertical and lateral resolution, making it a high-precision inversion approach.
2. Waveform-Guided Spatial Control
   Seismic data is often the only available information between wells, and lateral waveform variations reflect changes in lithofacies and seismic facies. Waveform-guided full-frequency inversion replaces the variogram-based spatial interpolation used in traditional geostatistics with lateral waveform variations, enabling automated seismic facies control. This overcomes the subjectivity of conventional facies-controlled inversion, which requires manual predefinition of depositional facies, making this method a truly seismic facies-constrained inversion technique.
3. Beyond Acoustic Impedance
   This method not only enables high-resolution inversion of acoustic impedance but also supports seismic-facies-constrained modeling of non-impedance parameters such as natural gamma, resistivity, and porosity. This represents a major advancement beyond the traditional limitation of seismic inversion to impedance results and significantly enhances the ability to simulate reservoir properties using seismic data.

Workflow of Full-Frequency Inversion:

The waveform-guided inversion sample selection process includes:

a. Constructing an isochronous stratigraphic framework;
b. Comparing predicted seismic waveforms with those adjacent to known wells and selecting the most similar ones as samples;
c. Analyzing vertical impedance structures of the sample wells and filtering high-frequency components using spectral decomposition to retain common deterministic features;
d. Using medium-frequency seismic impedance as a reference and optimizing the high-frequency components under a Bayesian framework;
e. Comparing synthetic and actual seismic impedance, retaining matches and discarding mismatches through repeated stochastic simulations;
f. Outputting the results as either expected values (averages of multiple realizations) or as a set of representative stochastic realizations.

Note: This method does not involve classifying seismic data into facies based on waveform shape, as classification standards are often ambiguous. Instead, known well waveforms are used as references. Samples are ranked based on waveform similarity and spatial proximity to ensure consistent structural features, enhancing spatial representation of depositional facies and improving alignment with sedimentary patterns. In practice, five or more wells within a 100 km^2 study area are typically sufficient for reliable results, even without uniform distribution.

Core Concept of Waveform-Guided Full frequency Inversion:

Waveform-guided inversion utilizes waveform similarity as an indicator and applies machine learning principles to drive broadband log simulation between wells, achieving high-resolution thin-layer inversion. The process includes:

1. Waveform Clustering

   This step involves applying singular value decomposition (SVD) for dynamic waveform clustering and establishing a mapping between waveform types and log structures. For instance, Figure 1(a) identifies two depositional facies types. Wells within the same facies show similar seismic waveforms and log characteristics. Based on waveform similarity, waveforms are grouped into two sample sets.
2. Initial Model Construction

   The next step is to conduct multi-scale analysis in the wavelet domain to identify common structural features among sample wells, which are used to construct the initial inversion model. For example, in Sample Set 1, the common log features from seven wells (represented by the curve in Figure 1) serve as the initial model.
3. Model Refinement Under Bayesian Framework

   The final process is to iteratively adjust the initial model based on actual seismic waveforms under a Bayesian framework until the inversion results match both the medium-frequency seismic data and the log structure features, resulting in a high-resolution, waveform-guided inversion.

Advantages and Methodological Innovations. Waveform-guided full-frequency inversion is a newly developed approach that significantly enhances both vertical and lateral resolution. It addresses the high-frequency randomness challenge in traditional geostatistical inversion and is particularly effective for predicting thin reservoirs with rapid lateral variations.

This method evolves from traditional geostatistics and adopts a **Seismic Waveform-guided Markov Chain Monte Carlo (SMCMC)** simulation algorithm. By leveraging waveform similarity, it identifies shared structural features within log data, ranks sample wells based on waveform similarity and spatial proximity, and selects those most related to the prediction point. High-frequency components are unbiasedly estimated, ensuring consistency between the final inversion results and the original seismic waveform.

The process involves:

- Analyzing known well locations and selecting those most correlated to the target trace;
- Using P-wave impedance from these samples as prior information;
- Matching the initial model with seismic band-limited impedance and computing a likelihood function;
- Taking advantage of the shared low-frequency characteristics across similar seismic waveforms;
- Combining likelihood and prior distributions in a Bayesian framework to derive posterior distributions;
- Perturbing model parameters to maximize posterior probability, and averaging multiple realizations as the final expected result.

Conceptual Framework: Replacing Variograms with Waveforms. The fundamental concept of waveform-guided full frequency inversion is to use lateral waveform variations under an isochronous framework to characterize reservoir spatial heterogeneity, instead of relying on variogram functions (Figure 1). Within the Bayesian framework, wells with high waveform similarity and spatial proximity are selected as samples to build the initial model. This enables well-seismic joint simulation with improved vertical and horizontal resolution for reservoir prediction.

Given the dense spatial distribution of seismic data and their strong correlation with depositional environments, waveform variations are more effective than variograms in representing the spatial distribution of sedimentary features, supporting more realistic facies-controlled stochastic inversion.

In traditional geostatistical inversion, variograms based on all available wells are used to optimize statistical samples based on spatial variation.

In waveform-guided full frequency inversion, statistical samples are selected based on both waveform similarity and spatial proximity, fully utilizing the lateral waveform variation.

Frequency Band Composition

The FFI spectrum comprises four frequency bands. Figure 2 shows schematic diagram of FFI principle:

- Low-Frequency (0–10 Hz): Derived from well logs and low-frequency models, capturing the overall structural trend and slow lateral variations. Velocity models contribute to constructing the low-frequency model with more reasonable spatial variability.
- Mid-Frequency (10–80 Hz): Extracted from seismic data, providing relative impedance information through relative acoustic impedance.
- High-Frequency (80–120 Hz): Estimated using waveform-facies-controlled inversion, which extend the inversion bandwidth and increase result determinism.
- Ultra-High-Frequency (>120 Hz): Estimated by combining well-log curves with seismic waveform using advanced intelligent cloud simulation technologies to improve well-log matching accuracy and prediction precision. incorporating waveform similarity and well-log statistics to estimate high-resolution attributes. The key challenge lies in ensuring that the simulated high-frequency content between wells is both realistic and geologically reasonable.

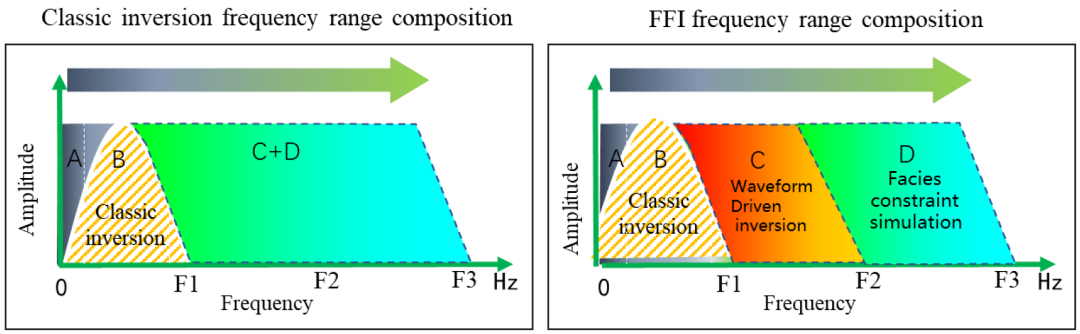


Figure 2—Schematic diagram of FFI principle

Shaped Spectrum Integration

The final full-frequency impedance volume is calculated using a shaped spectrum synthesis:

FFI=a*m*(F1)+b*n*(F2)+c*j*(F3)+d*k*(F4)

Where:

- FFI is the full-frequency composite spectrum;
- F1, F2, F3, F4 are the inversion results from the four frequency bands;
- a, b, c, d are weighting coefficients;
- m, n, j, k are spectrum shaping operators (e.g., triangular filters).

The principle of FFI is based on high-resolution inversion technology driven by seismic waveforms. It makes full use of seismic waveforms to reflect the microfacies change characteristics of the reservoir space, then analyzes the vertical combination structure of the reservoir. FFI inversion breaks through the limitation of 1/4 wavelength, realizes the fine prediction of thin layers in the vertical direction with a thickness up to 2m and achieves a qualitative leap in the accuracy of reservoir prediction. FFI includes the following four frequency band information.

First: low-frequency band (A) model, by interpolation of seismic velocity and well curves, leading to more reasonable spatial variations.

Second: medium-frequency band (B) information, from deterministic inversion.

Third: sub-high frequency band (C) information, using seismic waveform-driven facies-controlled reservoir prediction technology, using seismic waveforms as a tool to improve the certainty of inversion results. As figure 1 shows the seismic waveform is compared and analyzed with the well log bypass seismic waveform. The wells with similar waveform characteristics are selected as effective samples to construct a likelihood function that conforms to the seismic medium-frequency impedance and well curve structure characteristics. Figure 1 shows the basic schematic diagram of seismic waveform driven inversion.

Fourth: ultra-high frequency (D), using intelligent cloud simulation technology to increase the well compliance rate, improve vertical & horizontal resolution and prediction accuracy.

Fifth, Shaped weighted synthetic spectrum: FFI=a*m*(F1)+b*n*(F2)+c*j*(F3)+d*k*(F4).

FFI: Full-band synthetic spectrum, F1, F2, F3, F4 are the inversion results of four frequency bands; m, n, j, k: spectrum shaping operators, a, b, c, d: weight coefficients. Figure 2 shows the schematic diagram of full-frequency inversion principle.

Sixth, the full-frequency inversion method can not only realize high-resolution impedance inversion, but also realize facies-controlled high-resolution simulation of non-impedance parameters such as gamma, resistivity and porosity, breaking through the limitation that seismic inversion can only obtain impedance results. It is a great progress in using seismic information to simulate reservoir parameters.

Application to Carbonate Tidal Channel Reservoirs

The FFI method was applied to predict the internal structure of tidal channel reservoirs in the A units of the target formation.

Petrophysical Classification

Multi-well lithofacies intersection analysis shows a strong correlation between reservoir lithology and the porosity, resistivity and GR curves. By applying deep learning to explore the relationship between these curves and lithofacies, a standard and reference chart for lithofacies classification can be established.

Based on porosity, permeability, and electrical properties, three lithofacies were identified: Figure 3 shows that the target layer is divided into three rock types according to the porosity, permeability and electrical characteristics of the rock. The target unit has three types of lithofacies, namely high to middle permeability layer reservoir, middle permeability reservoir, and non-reservoir. It can be seen that a good progradation reservoir is characterized by high porosity, high resistivity and low gamma.

- High-permeability progradation reservoirs, the permeability more than 10mD.
- Medium- to low-permeability progradation reservoirs, permeability is 1 to 10mD.
- Non-reservoir facies, the permeability less than 1mD.

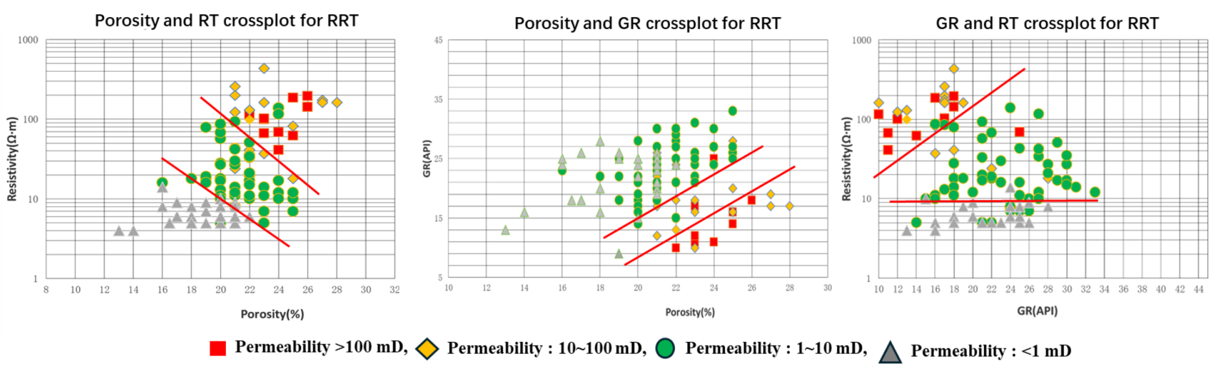


Figure 3—Three rock type

Petrophysical analysis revealed that the dominant pore types were moldic and intergranular pores. Log responses, with the best reservoirs concentrated in the progradation unit. Figure 3 shows seismic-well log facies classification based on porosity, permeability, and electrical properties.

Inversion application results

The study area is located in the northern block with the UAE. The sedimentary microfacies of the M layer are complex carbonate inner ramp. Multi-stage progradations are overlaid, the high permeability layers and barriers change frequently vertically, with fast lateral pinching out, and strong reservoir heterogeneity. Figure 4 shows the plane characteristics of the progradation seismic amplitude. The weak amplitude in the picture represents progradations.

In this paper, the FFI method is used to predict the combined structure and distribution characteristic of various reservoirs and hidden barriers in the strong heterogeneous carbonate reservoir in three-dimensional space.

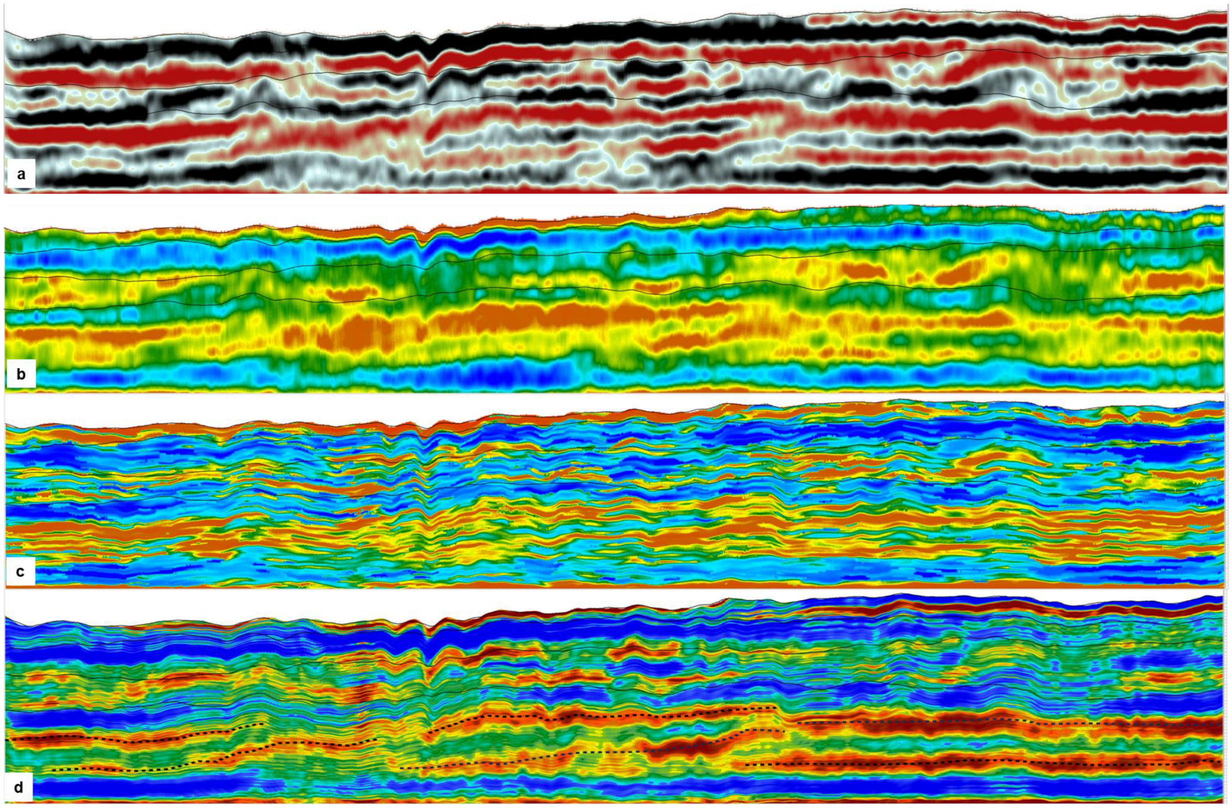


Figure 4—a. Seismic profile, b. CSSI, c. GSI, d. FFI, the red and yellow is reservoir, blue is no-reservoir

Secondly, FFI, CSSI, and GSI techniques were used to perform post-stack inversion on the M layer. The results showed that the progradation reservoir in A unit can be clearly predicted. The circled area in Figure 4a is the predicted progradation section. The good progradation reservoir shows low amplitude characteristics on the seismic section and low P-impedance characteristics on the inversion section by FFI. Figures 4b, c, and d circle the impedance sections of tidal channel reservoirs predicted by CSSI, GSI and FFI respectively. The application of FFI can clearly predict the spatial distribution characteristics of progradations and more clearly reflect the characteristics of geological sedimentary patterns in the vertical and lateral directions.

Thirdly, FFI was used to realize facies-controlled high-resolution simulation of resistivity. High-permeability layers exist in the progradation reservoirs in the study area, which are characterized by high porosity, high permeability and high resistivity. The full-frequency inversion not only can achieve high-resolution P-impedance inversion, but also can realize facies-controlled high-resolution simulation of non-P-impedance parameters such as gamma, resistivity and porosity, breaking through the limitation of seismic inversion that can only obtain P-impedance results. Since the high-permeability layers in carbonates are crucial for water injection development, FFI is used to realize facies-controlled high-resolution simulation of resistivity. The high-resolution prediction of the reservoir in the high permeability layers of the progradation is realized by the intersection of P-impedance and resistivity cubes. The high-resolution prediction of non-reservoir is realized by the intersection of P-impedance and gamma volume. The example shows that the prediction results are in good agreement with the known geological and logging data.

Through the comparative analysis of full-frequency inversion and the conventional inversion method, Constrained sparse spike inversion (CSSI) and geostatistical seismic inversion (GSI) results, it is further proved that seismic waveform driven inversion has higher accuracy. Compared with the geostatistical inversion results, the vertical accuracy reaches the geostatistical inversion results, but the geostatistical inversion results are random, and the full-frequency inversion results are deterministic. At the same time, the calculation time of geostatistical inversion is relatively long, while the calculation time of full-frequency inversion is relatively short. The same area of about 400km^2 needs 5 days to calculate the geostatistical inversion, while the full-frequency inversion only takes 12 hours on the same computer.

The FFI-based reservoir prediction effectively delineates the progradational features of the target formation. Figure 4 shows an FFI-predicted multi-well acoustic impedance section, where red indicates low-impedance reservoir zones. The progradation trend—from bottom to top and northeast to southwest—is clearly visible, with dashed lines marking the spatial extent of progradational bodies. The red-to-yellow zones represent predicted progradational units, whose distribution and boundaries align well with well data.

The FFI prediction shows good agreement with well results for the target reservoir, with a vertical resolution of 2–3 meters. It accurately maps the distribution of progradational bodies, revealing three main stacked sets in the target interval. FFI results indicate three distinct reservoir units—Progradation 1, Progradation 2, and Progradation 3—with clear vertical stacking or migration patterns. The progradational features of shoals are well defined, with sharp boundaries, supporting the design of more efficient development strategies.

Full-frequency inversion is a new inversion method developed in recent years. It can improve the vertical and horizontal resolution of the inversion results at the same time, solve the problem of strong randomness of high-frequency components in geostatistical inversion, and is more suitable for the prediction of thin reservoirs with fast lateral changes. It has achieved good application results in the prediction of progradational thin reservoirs in carbonates in the Middle East.

Conclusions

a. Full-frequency seismic inversion (FFI) leverages waveform-driven clustering and facies-controlled modeling to achieve high-resolution impedance prediction. It provides a robust solution to thin-layer identification in complex carbonate progradations.
b. FFI improves both vertical and lateral resolution beyond CSSI methods. Its reliance on waveform features and geologically constrained clustering enables realistic simulation of thin reservoir architecture. Full-frequency inversion reduces the uncertainty associated with geostatistical inversion and is also more computationally efficient.
c. Application in a northern UAE carbonate field demonstrated that FFI could accurately map progradations, interbeds, and high-permeability zones, supporting more effective reservoir management and development planning.

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